

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background.

A COMPREHENSIVE COVERAGE

MODULE - 15

AIR POLLUTION - 2

FLY ASH

Fly ash is ejected by thermal power plants as by-product of coal burning.

Composition of Fly Ash

Fly ash particles are made up of compounds of

- Aluminium
- Oxides of iron
- Calcium
- Silica and magnesium and
- toxic heavy metals like
 - ✓ Lead
 - ✓ Arsenic
 - ✓ Cobalt and
 - ✓ Copper



ENVIRONMENTAL EFFECTS OF FLY ASH

- Fly ash pollutes air and water and causes heavy metal pollution in water bodies.
- Fly ash affects vegetation as a result of its direct deposition on leaf surfaces or indirectly through its deposition on soil.
- Fly ash in the air slowly settles on leaves and crops in fields in areas near to thermal power plants and lowers the plant yield.



COMMERCIAL USES OF FLY ASH

- Cement can be replaced by fly ash up to 35%, thus reducing the cost of construction, making roads, etc.
- Fly ash bricks are light in weight and offer high strength and durability.
- Fly ash is a better fill material for road embankments and in concrete roads.
- Abandoned mines can be filled up with fly ash.
- Fly Ash can be used in landfills.



POLICY DECISION OF THE GOVERNMENT

The Ministry of Environment and Forests has made it mandatory to use Fly Ash based products in all construction projects and road embankment works within 100 kms radius of Thermal Power Station and mine filling activities within 50 kms radius of Thermal Power Station.



सत्यमेव जयते



Ministry of Environment, Forest
& Climate Change

BLACK CARBON

- Black carbon consists of pure carbon particles.
- They range in size from 5nm to 30nm.
- It is formed through the incomplete combustion of fossil fuels, biofuel, and biomass.
- Black carbon causes respiratory diseases and it is also a carcinogen.
- Black carbon warms the Earth by absorbing sunlight and heating the atmosphere and by reducing albedo when deposited on snow and ice.

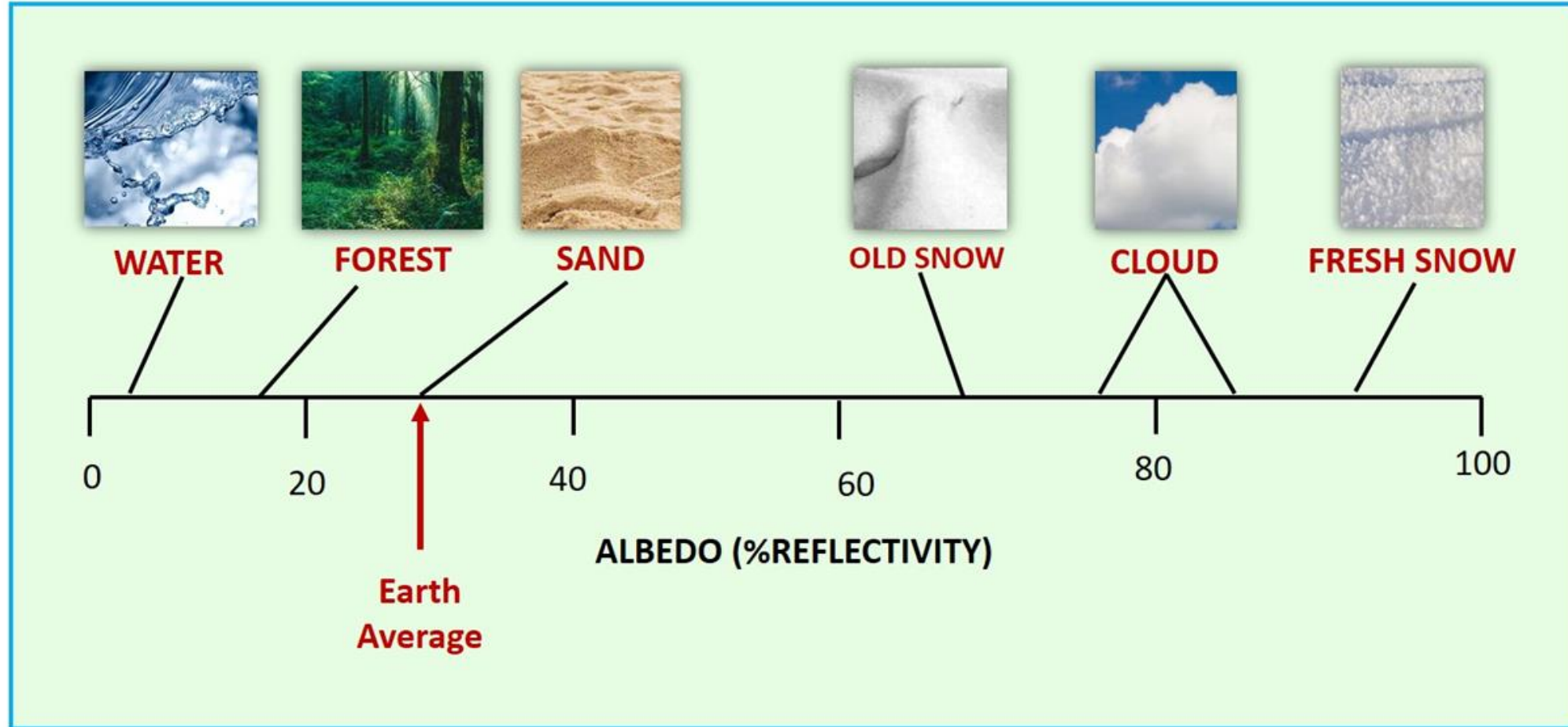


BLACK CARBON

- Black carbon stays in the atmosphere for only several days to weeks, whereas other greenhouse gases have longer lifecycles. Example, carbon dioxide has a lifetime of more than 100 years.
- Today, the majority of black carbon emissions are from developing countries and this trend is expected to increase.
- The largest sources of black carbon are Asia, Latin America, and Africa.
- China and India together account for 25-35% of global black carbon emissions. The majority of soot emissions in South Asia are due to biofuel cooking.



ALBEDO VALUES FOR EARTH SURFACES



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